

End Lab

01:18:29

100/100

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked.
[Learn more.](#)

[Open Google Cloud Console](#)

Username

student-02-48a409c3871b6



Password

k8qA6dWmw91T



GCP Project ID

qwiklabs-gcp-00-7121590



Create a Custom Search App using the Discovery API: Challenge Lab

< Previous

Next >

This lab may incorporate AI tools to support your learning.

Lab instructions and tasks



Challenge Lab Overview

Objective

Setup and requirements

Challenge scenario

Task 1. Create an AI Applications Data store and Search app

Task 2. Deploy a Python Website

Congratulations!

[< Previous](#)[Next >](#)

Challenge Lab Overview

This lab will challenge you to perform actions and automation across products. Instead of following step-by-step instructions, you are given a common business scenario and a set of tasks - you figure out how to complete them on your own! An automated scoring system (shown on this page) provides feedback on whether you have completed your tasks correctly.

When you take a Challenge Lab, you will not be taught Google Cloud concepts. You will need to use your skills to assess how to build the solution to the challenge presented. This lab is only recommended for students who have those skills. Are you up for the challenge?

Objective

[< Previous](#)[Next >](#)

Applications and Discovery API. You'll create a datastore with Python website URLs, build a search interface to query the data and display results.

This lab uses the following technologies and Google Cloud services:

- AI Applications
- Search Apps
- Agents
- Gemini

Setup and requirements

For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Make sure you signed into Qwiklabs using an **incognito window**.

[< Previous](#)

[Next >](#)

can finish in that time block.

There is no pause feature. You can restart if needed, but you have to start at the beginning.

3. When ready, click

START LAB

4. Note your lab credentials. You will use them to sign in to the Google Cloud

Open Google Console

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Username

google2876526_student@qwiklabs.n



Console.

[< Previous](#)

[Next >](#)

TG959yrKDX



GCP Project ID

qwiklabs-gcp-0855e773352d3560



[New to labs? View our introductory video!](#)

5. Click **Open Google Console**.

6. Click **Use another account** and copy/paste credentials for **this** lab into the prompts.

If you use other credentials, you'll get errors or **incur charges**.

7. Accept the terms and skip the recovery resource page.

Do not click **End Lab** unless you are finished with the lab or want to restart it. This clears your work and removes the project.

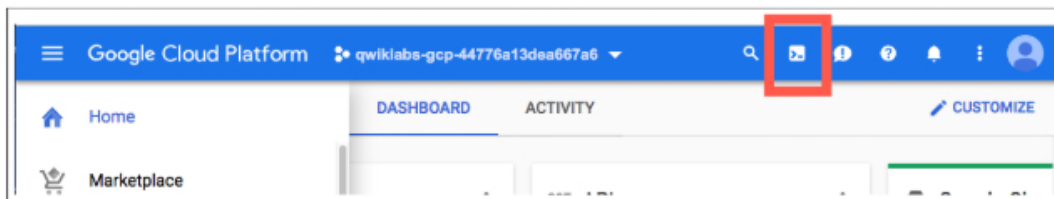
[< Previous](#)

[Next >](#)

Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

In the Cloud Console, in the top right toolbar, click the **Activate Cloud Shell** button.



Click **Continue**.

Cloud Shell

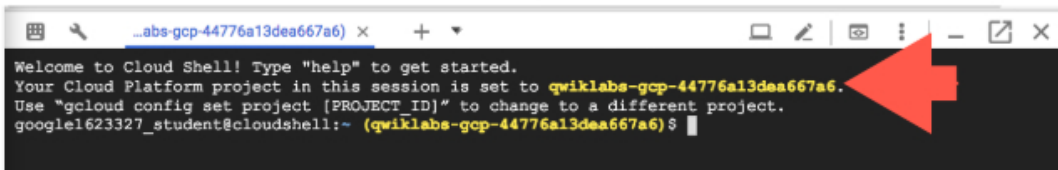
Google Cloud Shell provides you with command-line access to your cloud resources directly from your browser. You can easily manage your projects and resources without

[< Previous](#)

[Next >](#)

[Continue](#)

It takes a few moments to provision and connect to the environment. When you are connected, you are already authenticated, and the project is set to your *PROJECT_ID*. For example:



```
...abs-gcp-44776a13dea667a6) x + -  
Welcome to Cloud Shell! Type "help" to get started.  
Your Cloud Platform project in this session is set to qwiklabs-gcp-44776a13dea667a6.  
Use "gcloud config set project [PROJECT_ID]" to change to a different project.  
google1623327_student@cloudshell:~ (qwiklabs-gcp-44776a13dea667a6) $
```

`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

You can list the active account name with this command:

```
gcloud auth list
```

[< Previous](#)[Next >](#)

```
Credentialed accounts:
```

```
- <myaccount>@<mydomain>.com (active)
```

(Example output)

```
Credentialed accounts:
```

```
- google1623327_student@qwiklabs.net
```

You can list the project ID with this command:

```
gcloud config list project
```



(Output)

```
[core]
```

```
project = <project_ID>
```

[< Previous](#)[Next >](#)

```
[core]
```

```
project = qwiklabs-gcp-44776a13dea667a6
```

For full documentation of `gcloud` see the [gcloud command-line tool overview](#).

Challenge Scenario



< Previous

Next >

Cymbal Shops is an American retail chain headquartered in Minneapolis that sells homeware, electronics, and clothing.

Founded in 1974, Cymbal Shops started out as Cymbal Air, selling AC systems manufactured by Cymbal Group in Minnesota and neighboring states. The company quickly expanded into domestic merchandise in order to satisfy the need for quality products at an affordable price in the midst of a recession.

Cymbal Shops' broad product assortment was once a benefit - capturing planned purchases and last minute splurges. In recent years, however, the company has struggled to adapt to the acceleration in e-commerce. Digitally native companies are on the rise and Cymbal Shops must implement significant changes in order to keep pace and maintain relevance. Today, Cymbal Shops operates 714 stores across North America and reported \$15 billion in revenue in 2019. They currently employ 80,400 employees across the United States and Canada.

Cymbal Shops is a digitally transforming legacy retailer.

It is inspired by clients like: Bed Bath & Beyond, Best Buy, Home Depot,

< Previous

Next >

Your challenge

Cymbal Shops is looking for proficiency in creating a Custom Search App using the Discovery API. You work with Cymbal Shops, throughout the lab, you are tasked with deploying search app which involves two major tasks:

1. **Implementing Search Functionality:** Write a python code to connect with Discovery Engine, send user queries, and process results.
2. **Format search results:** Transform raw API data (titles, snippets, URLs) into a user-friendly display for the web application.

Task 1. Create an AI Applications Data store and Search app

[< Previous](#)[Next >](#)

In this section, you will utilize AI Applications to create a data store. Data stores act as knowledge bases, storing information from websites and documents that your agent can use to answer user questions. Therefore in this task,

1. Enable the [AI Applications API](#).
2. Create a data store that will be used by a generative agent to answer user questions. Configure the data store using the following details:

Property	Value
Data Store Name	cymbal-data-store
Data Source	Website Content
Website URLs	docs.python.org , www.w3schools.com/python
Location	global(Global)

Click **Check my progress** to verify the objectives.



Create an AI Applications Data store

Check my progress

3. Finally, create an application of type Search with the following configurations. This application will use the data store named `cymbal-data-store`.

Property	Value
App Type	Search
Content	Generic
Enterprise edition features	on
Advanced LLM features	on
Name the app	<code>cymbal-search-app</code>
Company Name	<code>cymbal-python-programming-company</code>
Location	global(Global)
Data Store	<code>cymbal-data-store</code>

4. Preview your app and ask it a question similar to: `What is the syntax of a Loop in Python?`

Note. It may take a short time for the Data Store to be available. If you get an error, wait a couple of minutes and try again.

Click **Check my progress** to verify the objectives.



Create a Search App

[Check my progress](#)

Assessment Completed!

Task 2. Deploy a Python Website

In this section, you will deploy a Python website. You can execute your Flask application directly within Cloud Shell, situated in the `app` directory, and preview it

by clicking the **Web Preview** button located at the top of the Cloud Shell editor.

< Previous

Next >

`vertex-search-assessment` folder.

```
gsutil cp -r gs://partner-genai-bucket/genai059/vertex-  
search-assessment/ .
```



2. You can further explore these files.

3. Setup the virtual environment, using the following commands.

```
pip install virtualenv  
virtualenv ~/genai-assessment  
source ~/genai-assessment/bin/activate
```



4. Populate the `requirements.txt` file with the Vertex AI module, and then install the packages listed in the `requirements.txt` file using `pip`.

```
echo "vertexai" >> requirements.txt
```



< Previous

Next >

5. In the `main.py` file, the code snippet outlines a Flask application that interacts with the Discovery Engine. Run the file using:

```
python main.py
```



Preview it on port 8080 by clicking on the **Web Preview** option located at the top-right of the Cloud Shell window.

If the deployment is successful, your application should look like this:

Output:

PyBot Home

PyBot

Python search:

[< Previous](#)[Next >](#)

Build a search interface for Google Discovery Engine Datastore

In the previous sub-section, you deployed a Python Flask application. Now, in this sub-section, you'll further develop the application by implementing the existing `search_data_store` and `format_response` functions within the `main.py` file. These functions are already defined but not yet implemented. Once implemented, they will enable users to search your Datastore and display formatted results.

1. In this task, your objective is to implement the `search_data_store` and `format_response` functions in the `main.py` file. There are provided comments within the code to guide you through the implementation.
2. The `search_data_store` function interacts with the Discovery Engine API to search your Datastore based on a given user query. It should return a list of `discoveryengine.SearchResponse` objects containing the search results.
3. The `format_response` function takes the raw search results obtained from `search_data_store` and transforms them into a suitable format for

[< Previous](#)[Next >](#)

The function should return a list of dictionaries. Each dictionary represents a formatted search result with the following keys:

- `title`: The title of the search result document.
 - `snippet`: A relevant snippet from the document.
 - `url`: The formatted URL of the search result document.
4. After making the necessary edits, run the application by executing the following command:

```
python main.py
```



Then, preview it on port 8080 by clicking on the **Web Preview** option located at the top-right of the Cloud Shell window.

Ask the **PyBot** application a sample question, such as: `What is the syntax of a Loop in Python?`

< Previous

Next >

PyBot Home

PyBot

Your Friendly Python Expert

Python search:

What is the syntax of a Loop in Python?

Submit

Python For Loops

Python For Loops. A for loop is used for iterating over a sequence (that is either a list, a tuple, a dictionary, a set, or a string).

https://www.w3schools.com/python/python_for_loops.asp

Python While Loops

With the while loop we can execute a set of **statements** as long as a condition is true. **Example**Get your own Python Server. Print i as long as i is less than 6: i ...

https://www.w3schools.com/python/python_while_loops.asp

4. More Control Flow Tools — Python 3.12.4 documentation

As well as the while **statement** just introduced, **Python** uses a few more that we will encounter in this chapter. If **Statements**: Perhaps the most well-known ...

<https://docs.python.org/3/tutorial/controlflow.html>

Congratulations!

At the end of this lab, you have gained practical experience in using AI Applications and the Discovery API to create a custom search solution.

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< Previous

Next >

